Contest Problem Set 12128

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Name _____

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1. $21 + 20 - 21 =$ _____.
2. The tens digit of 271828 is _____.
3. $591 + 324 =$ _____.
4. $56 \div 14 =$ _____.
5. The remainder of $149 \div 5$ is _____.
6. $24 + 37 + 13 + 25 =$ _____.
7. $432 - 276 =$ _____.
8. The product of 17 and 8 is _____.
9. $10 \times 19 \times 10 =$ _____.
10. (estimate) $1705 + 1023 + 682 =$ _____.
11. $89 \times 11 =$ _____.
12. $18^2 =$ _____.
13. $756 - 358 - 341 =$ _____.
14. The remainder of $469 \div 8$ is _____.
15. $34 \times 35 =$ _____.
16. *DCCCXLI* in Arabic numerals is _____.
17. $164 + 205 + 451 =$ _____.
18. $51 \times 49 =$ _____.
19. $18 \times 5 \times 21 =$ _____.
20. (estimate) $251 \times 498 =$ _____.
21. $24 \times 33 \div 9 =$ _____.
22. The GCD of 44 and 64 is _____.
23. $113 \times 25 =$ _____.
24. The greater of $\frac{11}{23}$ and $\frac{17}{35}$ is _____ (fraction).
25. $\frac{2}{5} + \frac{2}{15} + \frac{1}{45} =$ _____ (fraction).
26. $918 \div 17 =$ _____.
27. If 1760 yards is equal to 1 mile, then 220 yards is equal to _____ mile (fraction).
28. $32^2 =$ _____.
29. $24 + 29 + 34 + 39 + 44 + 49 =$ _____.
30. (estimate) $18 \times 202 \times 11 =$ _____.
31. $125^2 =$ _____.
32. The area of a rectangle with perimeter 24 and length 7 is _____.
33. $71 \times 99 =$ _____.
34. 288 hours is _____ days.
35. $66 \times 64 =$ _____.
36. The LCM of 44 and 64 is _____.
37. The number of multiples of 11 between 14 and 212 is _____.
38. 60% of 85 is _____.
39. The greatest prime divisor of 325 is _____.
40. (estimate) $\sqrt{314159} =$ _____.

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41. $61 \times 41 =$ _____.

42. The area of a right triangle with legs of length 28 and 16 is _____.

43. $65^2 - 34^2 =$ _____.

44. $18 \times 16 + 32 \times 6 =$ _____.

45. The remainder of $11674 \div 11$ is _____.

46. $\frac{26}{31} \times 26 =$ _____ (mixed number).

47. $2^4 \times 5^4$ _____.

48. The number of terms in the arithmetic sequence $7, 10, 13, \ldots, 109$ is _____.

49. $101 \times 491 =$ _____.

50. (estimate) $13 \times 285714 =$ _____.

51. The number of diagonals in a regular decagon is _____.

52. $\sqrt{13456} =$ _____.

53. $62 \times 222 =$ _____.

54. 2021_3 in base 9 is _____ $_9$.

55. $54\frac{6}{11}\%$ of 132 is _____.

56. The mean of the list $17, 25, 19, 31, 43$ is _____.

57. $118 \times 103 =$ _____.

58. The perimeter of a right triangle with legs of length 16 and 30 is _____.

59. $98 \times 79 =$ _____.

60. (estimate) $467 \times 749 =$ _____.

61. Three standard dice are rolled. The probability that the product of the numbers shown on the dice is odd is _____ (fraction).

62. $8\frac{3}{5} \times 12\frac{3}{5} =$ _____ (mixed number).

63. If $w = 38$, then $w^2 - 6w =$ _____.

64. $0.0\overline{36} =$ _____ (fraction).

65. In an isosceles triangle one angle measures 96°, and the measure of each of the other two angles is _____ $^\circ$.

66. $56 \times 375 =$ _____.

67. $13^3 =$ _____.

68. The number 265 written in base 8 is _____ $_8$.

69. $17 \times 16 \times 7 \times 6 =$ _____.

70. (estimate) $\frac{4}{3} \times 7^3 \times \pi =$ _____.

71. $972 \times 978 =$ _____.

72. If $3^{x+1} = 72$, then $9^{x-1} =$ _____.

73. $\sqrt{216} \times \sqrt{24} =$ _____.

74. The ninth term in the geometric series $\frac{3}{4}, \frac{3}{2}, 3, \ldots$ is _____.

75. $495^2 =$ _____.

76. The volume of a pyramid with a square base of side length 9 and a height of 4 is _____.

77. $625^{1.5} =$ _____.

78. The number of proper subsets of $\{P, R, I, Z, E\}$ is _____.

79. $63 \times 13 \times 11 =$ _____.

80. (estimate) $8.5 \times 6.5 \times 5.5 \times 9.5 =$ _____.

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Select E only if you cannot determine a uniquely correct answer between A, B, C, and D.

1.	A	B	C	D	E	11.	A	B	C	D	E	21.	A	B	C	D	E
2.	A	B	C	D	E	12.	A	B	C	D	E	22.	A	B	C	D	E
3.	A	B	C	D	E	13.	A	B	C	D	E	23.	A	B	C	D	E
4.	A	B	C	D	E	14.	A	B	C	D	E	24.	A	B	C	D	E
5.	A	B	C	D	E	15.	A	B	C	D	E	25.	A	B	C	D	E
6.	A	B	C	D	E	16.	A	B	C	D	E	26.	A	B	C	D	E
7.	A	B	C	D	E	17.	A	B	C	D	E	27.	A	B	C	D	E
8.	A	B	C	D	E	18.	A	B	C	D	E	28.	A	B	C	D	E
9.	A	B	C	D	E	19.	A	B	C	D	E	29.	A	B	C	D	E
10.	A	B	C	D	E	20.	A	B	C	D	E	30.	A	B	C	D	E

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1. What is the tens digit of the product $8 \times 9 \times 10 \times 11 \times 12$?

(A) 4 (B) 6 (C) 8 (D) 0 (E) Other

2. Vijay walks to and from school each day. The total distance he walks each day is two miles. If Vijay attends school every day of the week from Monday through Friday for four weeks, how many total miles does Vijay walk on his trips to and from school?

(A) 32 (B) 50 (C) 40 (D) 36 (E) Other

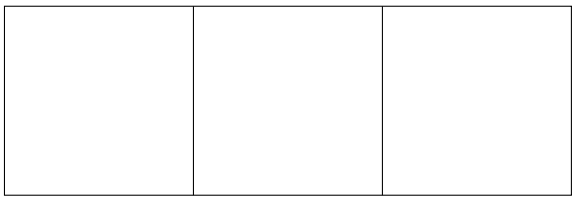
3. What number must replace all three $\square$'s in the equation $\square 7 + 9\square = 1\square 2$ in order to make the equation correct?

(A) 5 (B) 3 (C) 4 (D) 6 (E) Other

4. At the basketball game, Camille purchased four sodas for \$2 each, and three hot dogs for \$3 each. She paid for her entire purchase with a twenty dollar bill. How many dollars did Camille receive in return from the cashier?

(A) \$1 (B) \$4 (C) \$5 (D) \$2 (E) Other

5. The figure below shows a rectangle partitioned into three identical squares. How many rectangles or squares, of any size, can be drawn by tracing some or all of the figure?



(A) 6 (B) 4 (C) 3 (D) 5 (E) Other

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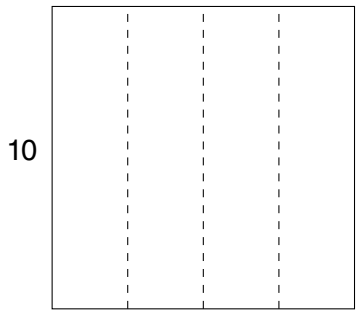
6. What is the least positive whole number that has a remainder of 1 when divided by 5 and a remainder of 3 when divided by 7?

(A) 16 (B) 26 (C) 36 (D) 31 (E) Other

7. On Monday, Jeff earned \$5 for helping his parents with housework. Each day after Monday, Jeff earned \$2 more than he did the previous day. How many total dollars had Jeff earned by the end of the day on Friday?

(A) \$34 (B) \$42 (C) \$60 (D) \$45 (E) Other

8. As shown below, a square of side length 10 is cut into four rectangles by three cuts parallel to two sides of the square. What is the total perimeter of these four rectangles?



(A) 120 (B) 60 (C) 80 (D) 100 (E) Other

9. Between 7 PM and 9 PM this evening, Joey watched three videos on ZooTube. The first video was 21 minutes and 38 seconds long. The second video was 12 minutes and 51 seconds long. The third video was 8 minutes and 26 seconds long. To the nearest minute, between 7 PM and 9 PM, how many minutes did Joey spend not watching videos on ZooTube?

(A) 77 (B) 79 (C) 75 (D) 78 (E) Other

10. What is the decimal value of $\frac{1}{5} + \frac{1}{25} + \frac{1}{125} + \frac{1}{625}$?

(A) 0.2496 (B) 0.2468 (C) 0.2486 (D) 0.24816 (E) Other

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11. Kenneth, Joanna, and Isaac are going to drive to a concert in a car that has driver's seat and a passenger seat in the front row, and a left and a right seat the back row. One of the three must sit in the driver's seat, which is in the front row. Isaac and Kenneth refuse to sit next to each other in the back row. How many different ways can the Kenneth, Joanna, and Isaac sit in the car for the ride to the concert?

(A) 18 (B) 12 (C) 10 (D) 24 (E) Other

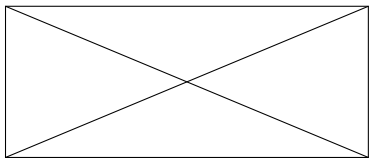
12. What is the remainder when $13 \times 17 \times 19 \times 23$ is divided by 12?

(A) 1 (B) 3 (C) 7 (D) 5 (E) Other

13. At Highville Park Middle School, 40% of the students in the Math Club are in the History Club, and 70% of the students in the History Club are in the Math Club. The History Club has 20 members. How many students are in the Math Club, but not the History Club?

(A) 18 (B) 35 (C) 28 (D) 21 (E) Other

14. A rectangle has a perimeter of 34 and an area of 60. What is the product of the lengths of the diagonals of the rectangle?



(A) 175 (B) 169 (C) 160 (D) 170 (E) Other

15. Sunil vacuums his apartment every fifth day, does the dishes every other day, and takes his dog for a long walk every third day. If today is January 1st and Sunil vacuums, does the dishs, and takes his dog for a long walk today, how many days this year, including today, will Sunil do all three tasks on the same day?

(A) 13 (B) 12 (C) 18 (D) 11 (E) Other

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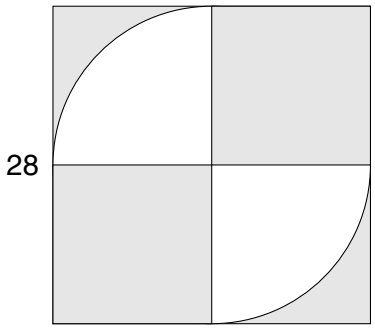
16. Chance rolls two standard six-sided dice. What is the probability that the product of the two numbers shown is a multiple of six?

(A) $\frac{5}{12}$ (B) $\frac{1}{2}$ (C) $\frac{4}{9}$ (D) $\frac{1}{3}$ (E) Other

17. Janna boarded the bus to visit the art museum downtown. At the first stop after she boarded the bus, three passengers exited the bus, and five more boarded the bus. At the second stop after she boarded the bus, one-third of the passengers exited the bus, and then a number of passengers boarded so that the number of passengers on the bus increased by 25%. At the third stop after Janna boarded the bus, half of the passengers exited, and then Janna exited as well, walking the rest of the way to the art museum. After Janna exited, there were four passengers on the bus. How many passengers were on the bus right before Janna boarded the bus?

(A) 11 (B) 8 (C) 10 (D) 9 (E) Other

18. Inside a gray square of side length 28, two white quarter circles are painted so that the corner of each quarter circle lies at the center of the square, and each quarter circle just touches two sides of the square. To the nearest whole number, what is the total area of the gray region?



(A) 498 (B) 476 (C) 528 (D) 422 (E) Other

19. Joshua left home to walk to the pier to catch the ferry. Eight minutes after he left home, Joshua was two-thirds of the way to the pier when he realized he would be late to catch the ferry, so he sprinted the rest of the way. Joshua sprints four times as fast as he walks. How long, in minutes, was Joshua's entire trip from home to the pier?

(A) 9.5 (B) 10 (C) 9 (D) 10.5 (E) Other

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20. If $a \spadesuit b = ab - 2a - 2b$, then what is the value of $(25 \spadesuit 4) \spadesuit (14 \spadesuit 5)$?

(A) 1204 (B) 1200 (C) 1196 (D) 1188 (E) Other

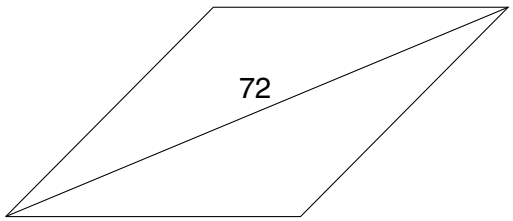
21. Over his most recent five basketball games, Luka scored a positive whole number of points each game, and the mean, median, and unique mode for Luka's five point totals were all 27 points. If the range of Luka's five point totals was 11 points, what is the greatest possible value for the number of points Luka scored in any of the five games?

(A) 33 (B) 34 (C) 36 (D) 32 (E) Other

22. What is the second greatest number in the list $\sqrt[6]{2021}$, 3.5, $\sqrt[3]{45}$, $\sqrt{13}$, $7^{\frac{2}{3}}$?

(A) $7^{\frac{2}{3}}$ (B) $\sqrt[3]{45}$ (C) $\sqrt[6]{2021}$ (D) 3.5 (E) Other

23. A rhombus has an area of 1080 and one diagonal of length 72. What is the perimeter of the rhombus?



(A) 136 (B) 144 (C) 156 (D) 168 (E) Other

24. Selena converted a two-digit positive whole number in base ten to base seven. The digits of the two-digit number in base seven were the same digits as the two-digit number in base ten, but in reverse order. What is the base ten sum of all of the base ten numbers that Selena could have converted?

(A) 74 (B) 56 (C) 69 (D) 72 (E) Other

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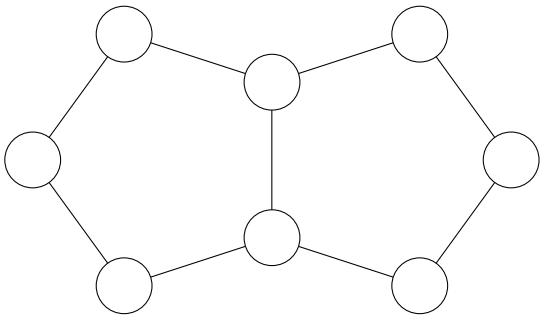
25. The diagram below shows two pentagons that share a common side. Circles are placed at each vertex, and each of the numbers from 1 through 8 is placed into one of the circles so that the sum of the five numbers at the vertices of either pentagon is the same. What is the greatest possible value of this sum?

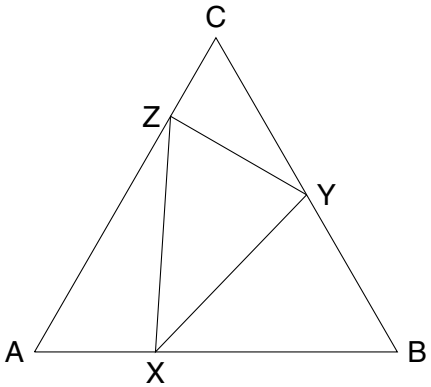


(A) 27 (B) 25 (C) 22 (D) 23 (E) Other

26. How many three-digit positive whole numbers have three distinct digits and are divisible by eleven?

(A) 60 (B) 62 (C) 65 (D) 64 (E) Other

27. In equilateral triangle *ABC*, shown below, point *Y* is the midpoint of side *BC*. Additionally, point *X* lies on side *AB* such that the length of *XB* is twice the length of *AX*, and point *Z* lies on side *AC* such that the length of *AZ* is three times the length of *CZ*. What is the ratio of the area of triangle *XYZ* to the area of triangle *ABC*?



(A) $\frac{\sqrt{2}}{5}$ (B) $\frac{\sqrt{3}}{6}$ (C) $\frac{1}{4}$ (D) $\frac{3}{11}$ (E) Other

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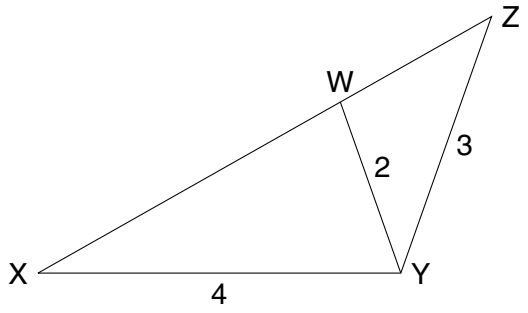
28. Jerry and Elaine both lost their smartphones, but both remembered they had lunch plans today. They each recalled a plan to meet at Monk's Cafe between noon and 1 PM, but neither could remember the exact time. Jerry plans to arrive at Monk's Cafe at a random time between noon and 1 PM, and will wait either ten minutes or until 1 PM, whichever comes first, for Elaine to arrive before leaving. Elaine is not quite as patient as Jerry. She plans to arrive at Monk's Cafe at a random time between noon and 1 PM, and will wait five minutes or until 1 PM, whichever comes first, for Jerry to arrive before leaving. The probability that Jerry and Elaine successfully meet for lunch is a common fraction. What is the sum of the numerator and denominator of that fraction?

(A) 355 (B) 370 (C) 332 (D) 308 (E) Other

29. In triangle *XYZ*, side *XY* has length 4 and side *YZ* has length 3. Point *W* lies on side *XZ* so that the length of segment *YW* is 2, and the length of *XW* is twice the length of *WZ*. What is the square of the length of *XZ*?



(A) 33 (B) 31 (C) 34 (D) 32 (E) Other

30. What is the sum of all positive whole numbers n for which the value of $\frac{n^2-2021}{n+1}$ also a positive whole number?

(A) 2223 (B) 4236 (C) 4242 (D) 3732 (E) Other

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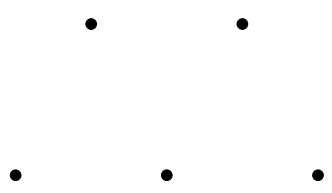
Problems 1 & 2

1. Today, Anna is twelve centimeters taller than her younger sister Julie. In the past year, Anna's height increased by five centimeters, and Julie's height increased by nine centimeters. One year ago, how much taller, in centimeters, was Anna than Julie?

1.

2. Five dots are shown below. Two dots are on the top row, and three dots are on the bottom row. The three dots on the bottom row are in a straight line. How many different triangles can be formed by choosing any three of the five dots as corners of the triangle?

2.



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Problems 3 & 4

3. Trey is trading candies with friends. He can trade four caramel candies for five butterscotch candies. He can also trade three butterscotch candies for two chocolate candies. Trey only has twelve caramel candies. Trey trades all of his caramel candies for butterscotch candies. He then trades all of his butterscotch candies for chocolate candies. How many chocolate candies does Trey have?

3.

4. How many positive whole numbers evenly divide both 168 and 192, but do not evenly divide 150?

4.

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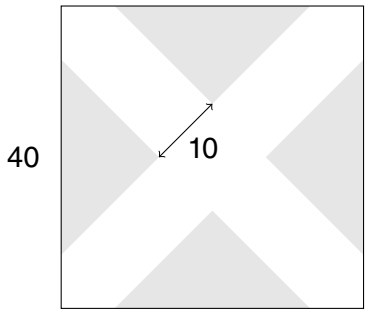
Problems 5 & 6

5. On a mathleague.org Sprint Round, there are 30 questions. Each correct answer is worth 4 points, each incorrect answer is worth -1 point, and an unanswered question is worth 0 points. At the most recent contest, Anish scored a 68 on the Sprint Round. What is the greatest possible number of questions Anish could have answered, either correctly or incorrectly, on the Sprint Round?

5.

6. As shown below, a gray square of side length 40 is crossed by two white stripes of width 10. Each stripe is parallel to a diagonal of the square, and the four gray regions created are all congruent. To the nearest whole number, what is the area of the region within the square that is white?

6.



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Problems 7 & 8

7. Daniel has a fair four-sided die, and each face on the die shows a unique number from 1 through 4. Daniel rolls the die three times. How many different ways can the numbers shown on the three rolls have a sum of eight?

7.

8. Lara, Mike, and Nikki each chose a positive whole number. There was no positive whole number other than 1 that evenly divided all of the numbers they chose, and 12 was the least positive whole number that was evenly divisible by all of the numbers they chose. How many different choices of numbers could Lara, Mike, and Nikki have made?

8.

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Team Round 12128

School or Team

Name _____

Name _____

Name _____

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1.	2.	3.	4.	5.
6.	7.	8.	9.	10.

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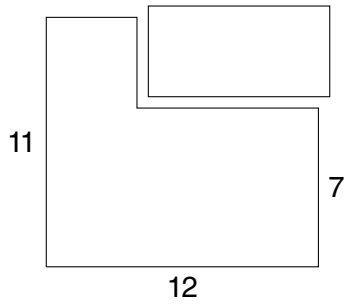
1. Catie was born in January of the year 2010. Including both the month she was born and May of 2021, how many months have passed from the month Catie was born until May of the year 2021?

2. Jalen placed each of the three digits 8, 9, and 2 into a $\square$ in the expression $\square\square \times \square$, and then calculated the product of the two-digit number and the one-digit number. For example, one such product is $29 \times 8 = 232$. What is the value of Jalen's largest product minus his smallest product?

3. Two cuts are made to a rectangular piece of paper measuring 12 inches by 11 inches, removing a smaller rectangular piece of paper, and leaving an L-shaped piece of paper. The area of the L-shaped piece of paper is 100, and one of the sides has a length of 7 inches, as shown below. What is the sum of the lengths of the two cuts, in inches?



4. April has two ten-dollar bills, three five-dollar bills, and fifteen one-dollar bills. If two bills are the same dollar amount, April cannot tell the bills apart from one another. How many different ways can April make a total of \$25? For example, one such way is one ten-dollar bill, two five-dollar bills, and five one-dollar bills.

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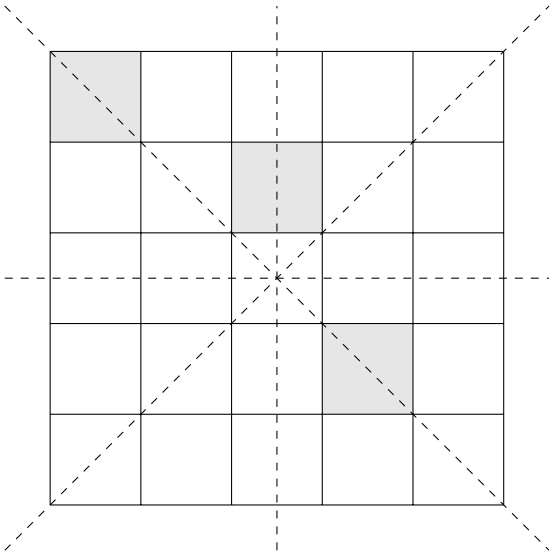
5. The diagram below shows a square partitioned into 25 smaller identical squares. Some of these squares are shaded. How many additional squares must be shaded so that when the square is folded along any of the dotted lines, a shaded square always coincides with a shaded square?



6. How many positive whole numbers less than or equal to 360 are divisible by 2 or 3, but are not divisible by 18?

7. In a standard deck of cards, there are thirteen ranks, and each rank has four cards. Nine of these ranks are whole numbers, ranging from 2 through 10. The other four ranks are not whole numbers. How many different pairs of cards, selected from a standard deck of cards, have ranks that are consecutive whole numbers? For example, one such pair would be the 7 of hearts and 6 of spades, since 6 and 7 are consecutive whole numbers.

8. Seven players are playing a game of chance. Each player is sitting in a chair numbered 1 through 7. In a bag are eight different chips. Seven of the chips are numbered 1 through 7, and an eighth chip is labeled *WINNER*. On each turn of the game, a player reaches into the bag and selects a chip. If it is the chip labeled *WINNER*, then the player wins the game. If it is any other chip, the chip is returned to the bag and the bag is given to the player in the seat with the same number as the chip that was just selected, and it is then that player's turn to select a chip. Note that if a player selects the chip with their own seat number, the chip is returned to the bag and it is that player's turn again. The game continues in this fashion until a player selects the chip labeled *WINNER* on their turn. The game begins with player 7 selecting a chip from the bag. What is the probability that player 7 wins the game? Express your answer as a common fraction.

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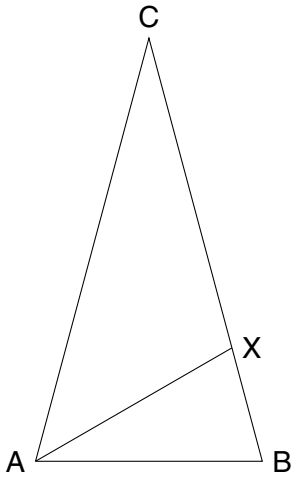
9. The nine numbered squares below are colored either red, green, or blue. Two of the squares are colored one color, three are colored a second color, and four are colored a third color. Additionally, all squares colored the same color must be immediately adjacent to one another. For example, squares 2 and 4 could not be the same color unless square 1 or square 5 is also that color. How many different ways can the nine squares be colored?

1	2	3
4	5	6
7	8	9

10. In isosceles triangle *ABC*, shown below, the two sides of equal length meet at vertex *C*, and the measure of angle *ACB* is 30 degrees. Point *X* is located on side *CB* such that the length of segment *AX* is equal to the length of segment *AB*. The area of triangle *ABC* is 1000. To the nearest whole number, what is the area of triangle *AXB*?



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Answers 12128

Sprint Round

1. A
2. C
3. A
4. E
5. A
6. D
7. D
8. D
9. A
10. A
11. E
12. A
13. D
14. B
15. A
16. A
17. D
18. B
19. C
20. C
21. B
22. E
23. C
24. C
25. B
26. D
27. E
28. A
29. A
30. B

Target Round

1. 16
2. 9
3. 10
4. 4
5. 27
6. 931
7. 12
8. 72

Team Round

1. 137
2. 560
3. 12
4. 8
5. 9
6. 220
7. 128
8. $\frac{1}{4}$
9. 288
10. 268

Number Sense

1. 20
2. 2
3. 915
4. 4
5. 4
6. 99
7. 156
8. 136
9. 1900
10. $[3240, 3580]$
11. 979
12. 324
13. 57
14. 5
15. 1190
16. 841
17. 820
18. 2499
19. 1890
20. $[118749, 131247]$
21. 88
22. 4
23. 2825
24. $\frac{17}{35}$
25. $\frac{5}{9}$
26. 54
27. $\frac{1}{8}$
28. 1024
29. 219
30. $[37997, 41995]$
31. 15625
32. 35
33. 7029
34. 12
35. 4224
36. 704
37. 18
38. 51
39. 13
40. $[533, 588]$
41. 2501
42. 224
43. 3069
44. 480
45. 3
46. $21\frac{25}{31}$
47. 10000
48. 35
49. 49591
50. $[3528568, 3899996]$
51. 35
52. 116
53. 13764
54. 67
55. 72
56. 27
57. 12154
58. 80
59. 7742
60. $[332294, 367272]$
61. $\frac{1}{8}$
62. $108\frac{9}{25}$
63. 1216
64. $\frac{2}{55}$
65. 42
66. 21000
67. 2197
68. 411
69. 11424
70. $[1365, 1508]$
71. 950616
72. 64
73. 72
74. 192
75. 245025
76. 108
77. 15625
78. 31
79. 9009
80. $[2743, 3031]$

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Solutions 12128

Sprint Round Solutions

1. As the product is a multiple of 10, the tens digit of the full product is the units digit of the product $8 \times 9 \times 11 \times 12$. The units digit of 8×9 is 2, and the units digit of 11×12 is also 2, and 2×2 has a units digit of $\boxed{4}$.

2. There are 5 days in each school week from Monday through Friday, so Vijay walks $5 \times 2 = 10$ miles each week. The total distance he walks, in miles, over 4 weeks is $4 \times 10 = \boxed{40}$.

3. The units digit is all that must be considered. The only digit that can be added to 7 to produce a units digit of 2 is $\boxed{5}$.

4. The total cost of the sodas was $4 \times \$2 = \$8$, and the total cost of the hot dogs was $3 \times \$3 = \$9$. The total cost of both the sodas and hot dogs was $\$9 + \$8 = \$17$, so the number of dollars received in return from the cashier was $\$20 - \$17 = \boxed{\$3}$.

5. The entire rectangle is 1 rectangle. Each square is 3 rectangles. Additionally, any two adjacent squares are 2 additional rectangles. Altogether, the number of rectangles is $1 + 3 + 2 = \boxed{6}$.

6. The numbers that have a remainder of 3 when divided by 7 are $3, 10, 17, 24, 31, 38, 45, \ldots$. Any number that has a remainder of 1 when divided by 5 must have a units digit of 1 or 6, and the least number that appears in the list with such a units digit is $\boxed{31}$.

7. There are 5 days from Monday through Friday, so the total amount Jeff earned was $\$5 + \$7 + \$9 + \$11 + \$13 = \boxed{\$45}$.

8. The perimeter of the square is $4 \times 10 = 40$. Each cut creates 2 sides of length 10. There are 3 such cuts, so the perimeter of the four rectangles exceeds the perimeter of the square by $3 \times 2 \times 10 = 60$. Thus, the total perimeter of the four rectangles is $40 + 60 = \boxed{100}$.

9. There are 2 hours from 7 PM until 9 PM, which is $2 \times 60 = 120$ minutes. The minute portions of the three videos total $21 + 12 + 8 = 41$ minutes. The second portions total $38 + 51 + 26 = 115$ seconds, which is very nearly 2 minutes. Thus, to the nearest minute, Joey spent $41 + 2 = 43$ minutes watching ZooTube videos, which means the number of minutes he spent not watching videos on ZooTube was $120 - 43 = \boxed{77}$.

10. This is $\frac{2000}{10000} + \frac{400}{10000} + \frac{80}{10000} + \frac{16}{10000}$, or $\frac{2496}{10000}$, which as a decimal is 0.2496.

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11. There are 3 possibilites for the driver. Then there are 3 possibilities for the passenger seat, including leaving it empty. Then there are 2 possibilities for the left back row seat, and only 1 possibility for the right back row seat. This is $3 \times 3 \times 2 \times 1 = 18$ possible seating arrangements. However, 2 of these have Joanna driving and Isaac and Kenneth in the back seat, and must be discarded. Therefore the total number of possible seating arrangements is $18 - 2 = \boxed{16}$.

12. This can be written as $(12+1) \times (12+5) \times (2 \times 12 - 5) \times (2 \times 12 - 1)$. When expanded, every term is a multiple of 12 except for $1 \times 5 \times (-5) \times (-1)$, which is 25. The remainder when 25 is divided by 12 is $\boxed{1}$.

13. There are 20 members of the History Club, so there are $0.7 \times 20 = 14$ members of the History Club that are also in the Math Club. Because 40% of the members of the Math Club are also in the History Club, there are $14 \div 0.4 = 35$ members of the Math Club. Therefore the number of students that are in the Math Club but not in the History Club is $35 - 14 = \boxed{21}$.

14. The sum of the length and width of the rectangle is $\frac{34}{2} = 17$, and the product of the length and width of the rectangle is 60. Considering factor pairs of 60, the values 5 and 12 have a sum of 17. By the Pythagorean theorem, the length of a diagonal of the rectangle is $\sqrt{5^2 + 12^2} = 13$, so the product of the lengths of the diagonals is $13^2 = \boxed{169}$.

15. Sunil vacuums every 5 days, does the dishes every 2 days, and walks the dog every 3 days. The first day he does all three is day 1 of the year. Because the least common multiple of 5, 2, and 3 is 30, the second day he does all three will be day $1 + 30 \cdot (2-1) = 31$ of the year. The final day he does all three will be day $1 + 30 \cdot (13-1) = 361$ of the year. The total number of days he does all three is $\boxed{13}$.

16. There are $6 \cdot 6 = 36$ total possible outcomes. If either die shows a 6, then the product will be a multiple of 6. This can occur in 11 different ways. Additionally, if one die shows a 3 and the other die shows a 2 or a 4, the product will be a multiple of 6. This can occur in 4 ways. Altogether, the total number of ways for the product to be a multiple of 6 is $11 + 4 = 15$, for a probability of $\frac{15}{36} = \boxed{\frac{5}{12}}$.

17. Working backwards, before Janna exited, there were $4+1 = 5$ passengers on the bus. Before half of the passengers exited, $5 \times 2 = 10$ passengers were on the bus. Increasing the number of passengers by 25% is multiplying the number of passengers by 1.25, so before those passengers boarded there were $\frac{10}{1.25} = 8$ passengers on the bus. One third of the passengers exiting the bus is multiplying the number of passengers by $1 - \frac{1}{3} = \frac{2}{3}$, so before those passengers exited there were $8 \div \frac{2}{3} = 12$ passengers on the bus. Before 5 passengers boarded and 3 passengers exited, there were $12 - (5-3) = 10$ passengers on the bus. Therefore before Janna boarded, the number of passengers on the bus was $10 - 1 = \boxed{9}$.

18. The area of the square is $28^2 = 784$. The two quarter circles together comprise a half circle of radius $\frac{28}{2} = 14$. The area of this half circle is $\frac{1}{2} \cdot 14^2 \cdot \pi$. Approximating π as $\frac{22}{7}$, this area is very nearly $\frac{1}{2} \cdot 14 \cdot 14 \cdot \frac{22}{7} = 7 \cdot 7 \cdot 22$, which is 308. The area of the gray region is, to the nearest whole number, $784 - 308 = \boxed{476}$.

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19. Joshua traveled $\frac{2}{3}$ of the way to the pier in 8 minutes, so if he continued walking the remaining $\frac{1}{3}$ of the trip would take 4 minutes. However, he sprints 4 times as fast as he walks, so the remaining $\frac{1}{3}$ of the trip only took $\frac{4}{4} = 1$ minute, and the number of minutes for Joshua's entire trip was $8 + 1 = \boxed{9}$.

20. The expression $ab-2a-2b$ can be rewritten as $(a-2)(b-2)-4$. Therefore $25\spadesuit 4$ is $(25-2)(4-2)-4 = 42$, and $14\spadesuit 5$ is $(14-2)(5-2)-4 = 32$. Finally, $42\spadesuit 32$ is $(42-2)(32-2)-4$, or $1200-4 = \boxed{1196}$.

21. Because it is the unique mode, at least two of Luka's point totals were 27. Let the greatest point total in the five games be x. Then because the range is 11, the least is $x - 11$. To maximize the value of x, the second least must be $x - 10$. Since the mean is also 27, the sum of all five scores is $27 \cdot 5 = 135$, so $(x - 11) + (x - 10) + 27 + 27 + x = 135$, and $3x + 33 = 135$. Solving for x, the greatest possible value for the number of points Luka scored in any of the five games is $\boxed{34}$.

22. The number 3.5 is less than $\sqrt{13}$ because $3.5^2 = 12.25$. Taking each number other than 3.5 to the power of 6 yields 2021, $45^2 = 2025$, $13^3 = 2197$, $7^4 = 2401$. Because 3.5 is less than $\sqrt{13}$ and 13^3 is greater than 45^2 and 2021, the second greatest number in the list is $\boxed{\sqrt{13}}$.

23. The area of a rhombus is half the product of the diagonals, so the second diagonal has length $\frac{2 \cdot 1080}{72} = 30$. The diagonals also bisect one another and partition the rhombus into four congruent right triangles. One of these right triangles has legs of length $\frac{30}{2} = 15$ and $\frac{72}{2} = 36$. The hypotenuse of this right triangle is a side of the rhombus. By the Pythagorean Theorem, this side length is $\sqrt{15^2 + 36^2} = 39$, so the perimeter of the rhombus is $4 \cdot 39 = \boxed{156}$.

24. Let the base ten number be $10a + b$. Since the base seven number has the same digits as the base ten number but in reverse order, we have $10a + b = 7b + a$. Then we use the properties of equality to get $2b = 3a$. Since $a \geq 1$ and $0 \leq a, b \leq 7$, we find that the base ten numbers are 23 and 46, and the sum is $\boxed{69}$.

25. To maximize the sum, the two numbers that are used twice should be as great as possible. If 7 and 8 are placed in the two circles that are used twice, the remaining numbers sum to $1+2+3+4+5+6 = 21$, which is odd and therefore cannot be evenly split between the three remaining circles for each pentagon. So the the maximum sum occurs when the two circles that are used twice have an 8 and a 6, splitting $1 + 2 + 3 + 4 + 5 + 7 = 22$ between the three remaining circles for each pentagon, so that each gets three numbers that sum to $\frac{22}{2} = 11$. Therefore the greatest possible value for the sum is $11 + 6 + 8 = \boxed{25}$.

26. Since 990 and 110 are the greatest and least three-digit multiples of 11, there are $\frac{990-110}{11} + 1 = 81$ three-digit multiples of 11. Now consider the numbers with two or more identical digits. There are 0 numbers where all three digits are identical, as 111 is not a multiple of 11 and $11 \cdot 111$ is not a three-digit number. When the hundreds and tens digits are identical, there are 9 numbers, namely every three-digit multiple of 110. When the tens and units digits are identical, there are 0 numbers, as the tens and units digits would form a two-digit number that is a multiple of 11 and no three-digit multiple of 100 is a multiple of 11. When the hundreds and units digits are identical, there are 8 numbers, as the hundreds and tens digits can be any value from 1 through 9 except for 5. Thus the number of three-digit multiples of 11 with three distinct digits is $81 - 9 - 8 = \boxed{64}$.

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27. Let the area of ABC be 1. Because Y is the midpoint of BC, the altitude from Y to side AB is $\frac{1}{2}$ the altitude of ABC. Additionally, the length of segment BX is $\frac{2}{3}$ the length of AB. Therefore the area of triangle XBY is $1 \cdot \frac{1}{2} \cdot \frac{2}{3} = \frac{1}{3}$. Similarly, the area of triangle XAZ is $1 \cdot \frac{1}{3} \cdot \frac{3}{4} = \frac{1}{4}$, and the area of triangle YCZ is $1 \cdot \frac{1}{4} \cdot \frac{1}{2} = \frac{1}{8}$. Therefore the area of triangle XYZ is $1 - \frac{1}{3} - \frac{1}{4} - \frac{1}{8} = \frac{7}{24}$, and since the area of ABC was assumed to be 1, the ratio of the area of XYZ to the area of ABC is also $\boxed{\frac{7}{24}}$.

28. If Jerry arrives at noon, Elaine can arrive any time between noon and 12:10. If Jerry arrives at 12:05, Elaine can arrive at any time between noon and 12:15. This 15 minute period for successfully meeting persists until Jerry arrives 12:50. Beyond 12:50, the window shrinks, so that if Jerry arrives at 12:55, Elaine can only arrive between 12:50 and 1 PM, and if Jerry arrives exactly at 1 PM, Elaine can only arrive between 12:55 and 1 PM. Plot this region on a 60×60 grid, with area 3600, so that the arrival time of Elaine is on the vertical axis and the arrival time of Jerry is on the horizontal axis. The upper unsuccessful region is a right triangle with vertices at $(0, 10)$, $(0, 60)$, and $(60, 50)$, which has an area of $\frac{1}{2} \cdot 50 \cdot 50 = 1250$. The lower unsuccessful region is a right triangle with vertices at $(5, 0)$, $(60, 0)$, and $(60, 55)$, which has an area of $\frac{1}{2} \cdot 55 \cdot 55 = \frac{3025}{2}$. The probability of not successfully meeting is $\frac{1250+1512.5}{3600}$, or $\frac{221}{288}$. Therefore the probability of successfully meeting is $1 - \frac{221}{288} = \frac{67}{288}$, and $67 + 288 = \boxed{355}$.

29. Extend side XY past Y to point P, so that segment ZP is parallel to segment WY. Then triangle WXY is similar to triangle ZXP, and the ratio of corresponding sides is $\frac{WX}{ZX}$, which is $\frac{WX}{WX+WZ} = \frac{2}{2+1}$, or $\frac{2}{3}$. Then the length of segment ZP is $\frac{3}{2} \cdot WY$, which is $\frac{3}{2} \cdot 2 = 3$. Additionally, the length of segment XP is $\frac{3}{2} \cdot XY$, which is $\frac{3}{2} \cdot 4 = 6$. Segment YP has length $6 - 4 = 2$. Triangle ZYP is isosceles. An altitude from Z to YP, and meeting YP at Q, partitions ZYP into two congruent right triangles, each with a hypotenuse of length 3 and a leg half the length of YP, or 1. By the Pythagorean Theorem, the altitude from Z to YP is $\sqrt{3^2 - 1^2} = 2\sqrt{2}$. Triangle XQZ is a right triangle with legs of length $2\sqrt{2}$ and $4 + 1 = 5$. The square of the length of XZ is $(2\sqrt{2})^2 + 5^2 = \boxed{33}$.

30. The given expression can be rewritten as $\frac{n^2-1}{n+1} + \frac{-2020}{n+1}$, or $n - 1 + \frac{-2020}{n+1}$. For this to be a positive whole number, the value of $\frac{-2020}{n+1}$ must be a positive whole number less than $n - 1$. Therefore $n + 1$ must be a divisor of 2020 that is greater than the square root of 2020. The prime factorization of 2020 is $2^2 \cdot 5^1 \cdot 101^1$, so 2020 has $(2+1)(1+1)(1+1) = 12$ divisors, of which 6 will be greater than the square root. These values for $n + 1$ are 2020, 1010, 505, 404, 202, and 101, so the values of n are 2019, 1009, 504, 403, 201, and 100, and their sum is $\boxed{4236}$.

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Target Round Solutions

1. Because Julie grew 9 centimeters and Anna only grew 4 centimeters, Julie is $9 - 5 = 4$ centimeters closer in height to Anna than she was a year ago. So one year ago, Anna was taller than Julie by a margin, in centimeters, of $12 + 4 = \boxed{16}$.

2. If the 2 dots on the top row are used to form a triangle, then 1 of the 3 dots from the bottom row must be used to complete the triangle, for a total of 3 triangles. If only 1 dot on the top row is used to form a triangle, then 2 of the 3 dots from the bottom row must be used to complete the triangle. There are 3 different pairs of 2 dots on the bottom row. Any of these 3 pairs of dots can be matched with 1 of the 2 dots on the top row, for $3 \times 2 = 6$ triangles. Altogether, the number of possible triangles is $3 + 6 = \boxed{9}$.

3. Trey has 12 caramel candies, which is $12 \div 4 = 3$ groups of 4 candies. If he trades these for butterscotch candies, he will recieve 3 groups of 5 butterscotch candies, or 15 butterscotch candies. This is 5 groups of 3 butterscotch candies. If he trades these for chocolate candies, he will receive 5 groups of 2 chocolate candies, for a total number of chocolate candies of $\boxed{10}$.

4. The prime factorization of 168 is $2^3 \cdot 3 \cdot 7$, and the prime factorization of 192 is $2^6 \cdot 3$. Any number that divides both must be a divisor of $2^3 \cdot 3$, as that is the largest set of primes common to both prime factorizations. The prime factorization of 150 is $2 \cdot 3 \cdot 5^2$. Therefore any number that has no more than 1 power of 2 and no more than 1 power of 3 will also divide 150. The only numbers that divide both 168 and 192 but not 150 are 2^2, 2^3, $2^2 \cdot 3$, and $2^3 \cdot 3$, for a total of $\boxed{4}$.

5. Since 68 is a multiple of 4, Anish could have answered $68 \div 4 = 17$ questions correctly and 0 questions incorrectly. However, this does not maximize the number of questions answered. To maintain the same score, Anish could answer 1 more question correctly, increasing his score by 4 points, and 4 more questions incorrectly, decreasing his score by $4 \times (-1) = -4$ points. Thus he could have answered $17 + 1 = 18$ correctly and 4 incorrectly, or $18 + 1 = 19$ correctly and 8 incorrectly and maintained a score of 68. Any greater number is not possible, as there are only 30 questions.Therefore the greatest number of questions Anish could have answered is $19 + 8 = \boxed{27}$.

6. The area of the square is $40^2 = 1600$. The length of an edge of the square that is white from a single stripe is the length of the leg of an isosceles right triangle with hypotenuse of length 10, or $\frac{10}{\sqrt{2}} = 5\sqrt{2}$. As there are two of these segments on each edge of the original square, the length of each side that is gray is $40 - 10\sqrt{2}$. The four gray regions can be combined into one square with this side length. The area of this square is $(40 - 10\sqrt{2})^2 = 1800 - 800\sqrt{2}$. Therefore the area of the white region is $1600 - (1800 - 800\sqrt{2}) = 800\sqrt{2} - 200$. To the nearest whole number, this is $\boxed{931}$.

7. If the smallest roll is a 1, then the other two rolls must be a 3 and a 4. The $1 - 3 - 4$ sequence can occur in $3! = 6$ different ways. If the smallest roll is a 2, then the other two rolls are either a 2 and a 4, or a 3 and a 3. Both the $2 - 2 - 4$ and $2 - 3 - 3$ sequences can occur in 3 different ways. Therefore the total number of different ways to get a sum of 8 is $6 + 3 + 3 = \boxed{12}$.

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8. The prime factorization of 12 is $2^2 \cdot 3^1$. Of the three numbers, at least 1 number must be divisible by 3, but no more than 2 may be divisible by 3, because if all three numbers were divisible by 3, then 1 would not be the only number that evenly divided all of the three numbers. There are 3 ways for exactly 1 of the numbers to be divisible by 3, and 3 ways for exactly 2 of the numbers to be divisible by 3. Thus the total number of ways to distribute the prime factor of 3 among the three numbers is $3 + 3 = 6$. For 2^2, at least 1 number must be divisible by 2^2, and no more than 2 numbers may be divisible by 2^2. As with the 3, this can occur in 6 ways. However, it is also possible for one number to be divisible by 2^2 and another number to be divisible by 2^1. This can occur in $3! = 6$ ways. Altogether, the number of ways to distribute the prime factors of 2^2 among the three numbers is $6 + 6 = 12$. The total number of ways to choose the three numbers is the total number of ways to distribute the 3 multiplied by the total number of ways to distribute the 2^2, as each pair of selections produces a unique set of three numbers. Thus, the total number of possible choices is $6 \cdot 12 = \boxed{72}$.

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Team Round Solutions

1. There are 12 months in one year, and January is the first month of the year. Catie was alive for all of the months in the years 2010 through 2020, which is 11 years, or $11 \times 12 = 132$ months. Additionally, May is the fifth month of the year, so Catie was alive for 5 months in the year 2021, for a total number of months of $132 + 5 = \boxed{137}$.

2. The largest product has the 2 in the units digit of the two-digit number, and $82 \times 9 = 738$, while $92 \times 8 = 736$, so the largest product is 738. The smallest product has the 2 as the single-digit number, and since 89 is less than 98, the smallest product is $89 \times 2 = 178$. The difference between these two products is $738 - 178 = \boxed{560}$.

3. The area of the original piece of paper was $11 \times 12 = 132$ square inches, so the area of the rectangle that was removed was $132 - 100 = 32$ square inches. One side of this rectangle has length $11 - 7 = 4$ inches, so the length of the other side is $32 \div 4 = 8$ inches. The sum of the length of the two cuts, in inches, is $8 + 4 = \boxed{12}$.

4. If April uses 2 tens, she can complete the remaining amount with 1 five or 0 fives and 5 ones, for 2 possibilities. If April uses 1 ten, she can complete the remaining amount with 3, 2, 1, or 0 fives and a necessary number of ones, for 4 possibilities. Finally, if April uses 0 tens, she can complete the remaining amount with 3 or 2 fives and a necessary number of ones, for 2 possibilities. Altogether, the total number of possibilities is $2 + 4 + 2 = \boxed{8}$.

5. Number the squares, with squares 1 through 5 left to right on the top row, 6 through 10 left to right on the next to top row, and so on. Because square 1 is shaded, folds along the horizontal, vertical, and diagonal dotted lines require squares 5, 21, and 25 to be shaded. Because square 8 is shaded, the fold along the horizontal line requires square 18 to be shaded, and when these squares are folded along the diagonal lines, squares 12 and 14 must be shaded as well. Finally, because square 19 is shaded, folds across the diagonal, vertical, and horizontal lines require squares 7, 9, and 17 to be shaded as well. Altogether, the total number of additional squares that must be shaded is $3 + 3 + 3 = \boxed{9}$.

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6. There are $360 \div 2 = 180$ numbers that are multiples of 2, and $360 \div 3 = 120$ multiples of 3. However, this double counts the multiples of $2 \times 3 = 6$, and there are $360 \div 6 = 60$ multiples of 6. There are $180 + 120 - 60 = 240$ numbers that are multiples of 2 or 3. Because 2 and 3 both divide 18, every multiple of 18 is also a multiple of 2 and 3, and is included among these 240 numbers. There are $360 \div 18 = 20$ numbers that are multiples of 18 that need to be discarded, so the number of positive whole numbers less than or equal to 360 that are divisible by 2 or 3 but not 18 is $240 - 20 = \boxed{220}$.

7. One of the two numbers must be a 3, 5, 7, or 9. There are 4 such odd ranks, and 4 cards in each rank, for $4 \cdot 4 = 16$ cards with odd ranks. For each of these cards, there are 2 even ranks that can create a pair of cards with consecutive numbers, and there are 4 cards in each rank, for $2 \cdot 4 = 8$ cards with even ranks that can create a pair of cards with consecutive numbers. Therefore, the total number of pairs of cards with ranks that are consecutive numbers is $16 \cdot 8 = \boxed{128}$.

8. There is a $\frac{1}{8}$ probability that player 7 wins the game on the first selection. There is a $\frac{7}{8}$ probability that player 7 does not win the game on the first draw, and, by symmetry, this $\frac{7}{8}$ probability is evenly distributed among all 7 players. Therefore player 7 has an additional $\frac{1}{7} \cdot \frac{7}{8} = \frac{1}{8}$ probability of winning after not winning on the first selection. Therefore the probability that player 7 wins the game is $\frac{1}{8} + \frac{1}{8} = \frac{1}{4}$.

9. Consider where the two squares the same color are placed. If they are in squares 5 and 8, or any other pair of squares that is 5 and an edge square, there are 2 ways to color the remaining squares. For example, squares 7, 4, 1, and 2 must all be the same color, or squares 9, 6, 3, and 2 must all be the same color, with the remaining squares being the remaining color. There are 4 ways to place the two squares that are the same color in 5 and an edge square, 2 ways to complete the coloring, and $3! = 6$ ways to permute the colors among a pattern, for a total of $4 \cdot 2 \cdot 6 = 48$ colorings in this case. The only other possibility for the two squares of the same color is to be in a corner square and an edge square. For example, squares 4 and 7. In this case, the four squares that are the same color must be in squares 1, 2, 3, and 6, or squares 1, 2, 3, and 5, or squares 1, 2, 5, and 8, or squares 3, 6, 9, and 8, or squares 5, 6, 8, and 9. In each case, the remaining three squares are the third color. There are 8 ways to choose the edge square and the corner square for two squares of the same color, then 5 ways to place the four squares that the same color, and $3! = 6$ ways to permute the colors among a pattern, for a total of $8 \cdot 5 \cdot 6 = 240$ colorings in this case. Altogether, the number of possible ways to color the squares is $48 + 240 = \boxed{288}$.

10. Because ACB is isosceles, the measure of angle CBA is $\frac{1}{2} \cdot (180^\circ - 30^\circ) = 75^\circ$. Because triangle AXB is isosceles, the measure of angle XAB is 30°, so the measure of angle CAX is $75^\circ - 30^\circ = 45^\circ$. Choose point Y on AC so that XY is perpendicular to AC. Then triangle AYX is an isosceles right triangle, and triangle CXY is a $30 - 60 - 90$ triangle. Let the length of XY be x. The length of AX, which is also the length of AB, is $x\sqrt{2}$. Additionally, the length of CY is $x\sqrt{3}$, and the length of AY is x, so the length of AC is $x + x\sqrt{3}$. Additionally, the length of CX is $2x$, so the length of XB is $x + x\sqrt{3} - 2x = x\sqrt{3} - x$. Triangle AXB is similar to triangle CAB, and the ratio of their sides is $\frac{XB}{AB} = \frac{\sqrt{3}-1}{\sqrt{2}}$. The ratio of their areas is the square of this value, so the area of AXB is $\left(\frac{\sqrt{3}-1}{\sqrt{2}}\right)^2 \cdot 1000$, which to the nearest whole number is $\boxed{268}$.